

AI News Intelligence Agent

Complete Project Guide & Interview Notes (Simple English Edition)

1. The Big Picture: What Problem Does This Project Solve?

Real-World Problem: Every morning, tech professionals and researchers read news from dozens of websites (TechCrunch, Reuters, Bloomberg, etc.). They face 3 major problems:

- 1. **Headline Overload:** Too many articles to read in full.
- 2. **Duplicate News:** 10 different websites report on the exact same press release.
- 3. **Superficial Content:** Articles are full of promotional filler and miss *why it matters*.

Our Solution: The **AI News Intelligence Agent** automates the entire news lifecycle. It fetches current news, uses a **Fine-Tuned Open-Source AI Model** to write crisp bulleted summaries, groups duplicate stories using vector math, and lets you ask questions directly to the news using **RAG (Retrieval-Augmented Generation)**.

2. Tech Stack & Why Each Technology Was Chosen

Technology	Role in Project	Why Chosen? (Simple English)
Streamlit	Frontend Dashboard	Allows building interactive web dashboards purely in Python without writing HTML/JS/CSS boilerplate.
PyTorch & Transformers	Local AI Model Engine	Industry standard open-source framework for running machine learning models locally.
PEFT (LoRA)	Model Fine-Tuning	Fine-tunes open-source LLMs by training only 0.4% adapter parameters, making training fast on standard PCs.
Sentence Transformers <i>(all-MiniLM-L6-v2)</i>	Text Embeddings	Converts text into 384-dimensional mathematical vectors to measure concept meaning and similarity.
FAISS	Vector Database Search	Facebook AI Similarity Search — performs sub-millisecond vector retrieval for grounded Q&A.;
Groq API	Cloud Fallback Engine	Acts as a fallback LLM if the user's local PC lacks GPU memory to run the model locally.

3. Core AI Concepts Explained in Simple Analogies

A. What is Fine-Tuning (LoRA)?

Analogy: Imagine hiring a general college graduate who speaks fluent English. They can write long essays, but they don't know how to write concise 3-bullet medical summaries. Fine-tuning is giving them specialized training.

What is LoRA? Instead of retraining their whole brain (costing millions of dollars), LoRA attaches a small specialized notebook (adapter weights). They consult the notebook to output perfectly formatted news summaries every time. We train only **0.4% parameters!**

B. What are Embeddings & Cosine Similarity?

Analogy: Embeddings map concepts on a 2D map. 'Apple' and 'Microsoft' are placed close together under 'Tech', while 'Banana' is far away under 'Fruit'.

Our model converts news articles into 384 numbers (vectors). **Cosine Similarity** measures the angle between two article vectors. If the angle is close to 0°, the articles report on the exact same news event!

C. What is RAG (Retrieval-Augmented Generation)?

Analogy: An Open-Book Exam.

Without RAG, if you ask an AI model about today's news, it guesses or invents facts (hallucination).

With RAG, when you ask a question, the FAISS vector database retrieves the top 3 relevant news articles and places them in front of the AI model. The AI reads those exact articles and answers your question with 100% grounded facts.

4. System Architecture Pipeline (How It Runs Step-by-Step)

1. **News Collection:** User enters a query -> System pulls headlines via Google News RSS & GNews API.
2. **Summarization:** Article content passes through local Fine-Tuned Model (LoRA) -> Structured summary.
3. **Embedding & Indexing:** Text is converted into vectors and indexed into FAISS vector database.
4. **Duplicate Detection:** Pairwise cosine similarity groups matching headlines across news outlets.
5. **RAG Q&A:** User asks a question -> FAISS retrieves context -> Model answers with source links.

5. How to Answer Key Interview Questions

Q: How did you build this project?

A: I built a modular Python Streamlit dashboard. I used HuggingFace PEFT to fine-tune an open-source model using LoRA for structured news summarization. I integrated Sentence Transformers and FAISS vector search for semantic duplicate detection and grounded RAG Q&A.;

Q: Why do you have both Fine-Tuning AND RAG? Aren't they redundant?

*A: No! Fine-tuning teaches the model **HOW** to format and structure news summaries. RAG gives the model **WHAT** real-time fresh news facts to read today. Fine-tuning handles format/style, while RAG handles up-to-date facts.*

Q: What happens if the local fine-tuned model cannot run on a weak PC?

A: The architecture uses a resilient multi-tier fallback mechanism. If the local PyTorch model fails to load, it automatically routes inference to Groq API cloud inference. If API keys are absent, it uses a clean heuristic fallback so the app never crashes.

6. Summary of Key Project Features

- **Live Multi-Source News Engine:** Pulls real-time headlines with geolocation support.
- **Domain-Specific LoRA Summarizer:** 3-bullet points + 'Why It Matters' impact analysis.
- **Semantic Duplicate Clustering:** Groups matching coverage across publishers.
- **Grounded RAG Q&A:** Interactive question answering with source citations.
- **Futuristic Dark UI:** Modern glassmorphism dashboard with real-time model status badges.